

Smart Home Automation System

Project Report

Overview

This project presents a Smart Home Automation System developed using an ESP32 microcontroller, relay modules, and various sensors. The system enables users to monitor and control household electrical appliances through both Bluetooth and Wi-Fi communication, providing a flexible and user-friendly home automation solution.

Objectives

- Automate household appliances.
- Provide wireless control using Bluetooth and Wi-Fi.
- Improve convenience, safety, and energy efficiency.

Hardware Used

- ESP32 Development Board
- Relay Module(s)
- Sensors (e.g., temperature, motion, or other project sensors)
- Power Supply
- Household electrical loads

Working Principle

The ESP32 acts as the central controller. Commands sent from a smartphone over Bluetooth or Wi-Fi are processed by the ESP32, which switches the connected relay modules ON or OFF to control appliances. Sensor data can also be read by the ESP32 to enable automatic actions based on environmental conditions.

Features

- Bluetooth-based appliance control
- Wi-Fi remote control
- Real-time switching using relay modules
- Sensor integration for automation
- Scalable design for additional devices

Applications

Home automation, energy management, smart offices, laboratories, and IoT learning projects.

Conclusion

The Smart Home Automation System demonstrates how the ESP32 can be used to build a low-cost, reliable, and expandable automation platform. By combining Bluetooth, Wi-Fi, relay modules, and sensors, the project provides an effective solution for controlling and monitoring home appliances.